

MANISH GUPTA

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SUMMARY

I am currently working with Google on MLIR-based code-generation for NVIDIA GPUs. I have completed Ph.D. in computer science from UC San Diego and Bachelors from IIT Roorkee.

EXPERIENCE

- **Software Engineer, Google, Mountain View, CA** Jun 2022 – Present
 - Improved half- (F16) and single-precision (F32) [OpenXLA](#) codegen performance targeting NVIDIA A100 Tensor core by 1.65x and 1.53x, respectively (see [slides](#), [talk](#))
 - Matched the performance for codegen-erated kernels with equivalent handwritten (CUTLASS) and library (cuBLAS) performance
 - Currently, working on code generation for NVIDIA Hopper H100 GPUs
- **Software Engineer, NVIDIA, Santa Clara, CA** Dec 2017 – May 2022
 - CUDA kernel development in [CUTLASS](#) to enable deep learning primitives on NVIDIA GPUs
 - Implemented convolution (forward and backward) for NVIDIA Ampere, Turing, and Volta architecture targeting CUDA and Tensor cores (see [CUTLASS 2.4 release](#))
 - Presented Implicit GEMM convolution at GTC 2021 (see [slides](#), [talk](#), and [notes](#))
 - Improved backward *strided* data gradient by 4x vs. to cuDNN 8.2 ([source code](#) and [notes](#))
 - Improved backward weight gradient through Split-k *serial* and *parallel* reductions for GEMMs and convolutions (see [notes](#))
 - Implemented complex GEMMs on floating-point (F32) data type targeting NVIDIA Ampere Tensor cores for Tensor Float 32 (TF32) data type ([source code](#) and [notes](#))
 - Implemented fast Gaussian complex GEMMs using 3m complex multiply algorithm targeting NVIDIA Ampere Tensor cores for double precision (F64) data type ([source code](#))
 - Led the implementation for CUTLASS profiler to ensure functional correctness and measure performance of deep learning operations ([source code](#))
 - Proposed and implemented LLVM-IR inspired representation of GEMM and Convolution operations ([source code](#))
- **Software Engineer, AMD Research, Austin, TX** Jan 2016 – Sep 2016
 - Compiler passes using LLVM for GPU kernels written in C++AMP
 - Reliability vs. performance trade-offs study for memories
 - Worked with the U.S. Department of Energy (DoE) on [Fast Forward exascale computing project](#)
- **Firmware Engineer, Qualcomm, Hyderabad, India** Sep 2008 – Aug 2010
 - Coded and integrated features such as Dolby Digital decoder, echo-canceller & noise suppressor, automatic gain control, and IIR filters on Qualcomm ADSP
 - Acquired hands-on knowledge of processor architecture, memory management, OS, assembly & C language, RTOS, firmware, system simulation, Tcl programming, and JTAGs
- **Research Assistant, UCSD, San Diego, CA** Sep 2010 – Sep 2017
 - Static taint analysis for CUDA kernels using LLVM
 - Software-based recovery mechanisms for code executing on unreliable hardware
 - Compiler infrastructure to analyze and generate reliable code

EDUCATION

- **University of California, San Diego (UCSD)**
 - Ph.D. in Computer Science and Engineering, Powell Fellow Sep 2010 – Sep 2017
 - M.S. in Computer Science and Engineering, GPA 3.8/4.0 Sep 2010 – Jun 2012
- **Indian Institute of Technology, Roorkee, India (IIT-R)**
 - B.Tech. in Electrical Engineering, Gold Medalist, GPA 9.2/10.0 May 2004 – Jun 2008

- **Software Techniques to Enhance Reliability of Memory and Compute Units**

Written by: Manish Gupta

Advised by: Prof. Rajesh Gupta and Prof. Dean Tullsen

University of California, San Diego, 2017

PUBLICATIONS (SELECTED 3 OUT OF 5)

- **Reliability-aware Data Placement for Heterogeneous Memory Architecture**

Manish Gupta, Vilas Sridharan, David Roberts, Andreas Prodromou, Ashish Venkat, Dean Tullsen and Rajesh Gupta in HPCA 2018

- **Compiler Techniques to Reduce the Synchronization Overhead of GPU RMT**

Manish Gupta, Daniel Lowell, John Kalamatianos, Steven Raasch, Vilas Sridharan, Dean Tullsen, Rajesh Gupta in DAC 2017

- **Verifying GPU Kernels by Test Amplification**

Alan Leung, Manish Gupta, Yuvraj Agarwal, Rajesh Gupta, Ranjit Jhala, and Sorin Lerner in PLDI 2012

PATENTS (SELECTED 2 OUT OF 10)

- **Bufferless Communication for RMT using Register Permutation** ([link](#))

- **Waterfall Counters with Application to AVF Estimation** ([link](#))

SKILLS

- **Programming Languages:** C++, C, Python, Shell, Java, MATLAB, Haskell

- **Parallel Programming:** CUDA, C++AMP, OpenCL

- **Simulators and Tools:** Ramulator, FaultSim, L^AT_EX, PinTool, Hadoop, Gem5, McPAT, QEMU

- **Benchmarks:** ResNet50, MaskRCNN, BERT, SPEC, PARSEC, Proxy Apps, HPC

TEACHING EXPERIENCE

- Basic Data Structure & Object Oriented Design, Teaching Assistant, Fall 2015

- Software for Embedded Systems, Teaching Assistant, Spring 2014

Please find my teaching reviews at links: [Teaching Eval 1](#) & [Teaching Eval 2](#)

LANGUAGES

- English (proficient), Hindi (native), and Russian (basic)